

Ex.no:2	EDA-DATA IMPORT AND EXPORT

Aim:

To inspect the features of the City Day Air Quality dataset, detect missing values, and export the dataset into CSV, Excel, and SQLite database formats using Python.

Algorithm:

1. Import pandas and sqlite3 libraries.
2. Read city_day.csv using pd.read_csv().
3. Inspect the dataset using .head(), .tail(), .info(), .describe(), and .isnull().sum().
4. Display the dataset shape and column names.
5. Export the DataFrame to CSV and Excel formats using .to_csv() and .to_excel().
6. Connect to a SQLite database and write the dataset into an SQL table using .to_sql().
7. Close the database connection and end execution.

Program:

```
# CITY DAY AIR QUALITY DATASET

import pandas as pd
import sqlite3

# Load the City Day dataset
df = pd.read_csv("city_day.csv")

# Display complete dataset
print("===== DATASET =====")
print(df)

# Display first five records
print("\n===== FIRST FIVE ROWS =====")
print(df.head())

# Display last five records
print("\n===== LAST FIVE ROWS =====")
print(df.tail())

# Display dataset information
print("\n===== DATASET INFORMATION =====")
df.info()
```

```
# Display statistical summary
print("\n===== STATISTICAL SUMMARY =====")
print(df.describe(include='all'))

# Display dataset shape
print("\nDataset Shape:", df.shape)

# Display column names
print("\n===== COLUMN NAMES =====")
print(df.columns.tolist())

# Check missing values
print("\n===== MISSING VALUES =====")
print(df.isnull().sum())

# Export dataset to CSV
df.to_csv("city_day_export.csv", index=False)
print("\nCSV file exported successfully!")

# Export dataset to Excel
df.to_excel("city_day_export.xlsx", index=False)
print("Excel file exported successfully!")

# Create SQLite database
conn = sqlite3.connect("city_day.db")

# Export DataFrame to SQL table
df.to_sql(
    "city_day",
    conn,
    if_exists="replace",
    index=False
)

print("Dataset exported successfully to SQLite database!")

# Close database connection
conn.close()

print("SQLite database connection closed successfully!")
```

Output:

```
===== DATASET =====
      City      Date  PM2.5  PM10    NO    NO2    NOx    NH3  \
0      Ahmedabad  2015-01-01    NaN    NaN    0.92  18.22  17.15    NaN
1      Ahmedabad  2015-01-02    NaN    NaN    0.97  15.69  16.46    NaN
2      Ahmedabad  2015-01-03    NaN    NaN   17.40  19.30  29.70    NaN
3      Ahmedabad  2015-01-04    NaN    NaN    1.70  18.48  17.97    NaN
4      Ahmedabad  2015-01-05    NaN    NaN   22.10  21.42  37.76    NaN
...      ...      ...      ...      ...      ...      ...      ...
29526  Visakhapatnam  2020-06-27   15.02  50.94    7.68  25.06  19.54  12.47
29527  Visakhapatnam  2020-06-28   24.38  74.09    3.42  26.06  16.53  11.99
29528  Visakhapatnam  2020-06-29   22.91  65.73    3.45  29.53  18.33  10.71
29529  Visakhapatnam  2020-06-30   16.64  49.97    4.05  29.26  18.80  10.03
29530  Visakhapatnam  2020-07-01   15.00  66.00    0.40  26.85  14.05   5.20

      CO      SO2      O3  Benzene  Toluene  Xylene  AQI  AQI_Bucket
0      0.92  27.64  133.36    0.00    0.02    0.00    NaN      NaN
1      0.97  24.55   34.06    3.68    5.50    3.77    NaN      NaN
2     17.40  29.07   30.70    6.80   16.40    2.25    NaN      NaN
3      1.70  18.59   36.08    4.43   10.14    1.00    NaN      NaN
4     22.10  39.33   39.31    7.01   18.89    2.78    NaN      NaN
...      ...      ...      ...      ...      ...      ...      ...
29526    0.47   8.55   23.30    2.24   12.07    0.73   41.0      Good
29527    0.52  12.72   30.14    0.74    2.21    0.38   70.0  Satisfactory
29528    0.48   8.42   30.96    0.01    0.01    0.00   68.0  Satisfactory
29529    0.52   9.84   28.30    0.00    0.00    0.00   54.0  Satisfactory
29530    0.59   2.10   17.05     NaN     NaN     NaN   50.0      Good

[29531 rows x 16 columns]

===== FIRST FIVE ROWS =====
      City      Date  PM2.5  PM10    NO    NO2    NOx    NH3    CO    SO2  \
0  Ahmedabad  2015-01-01    NaN    NaN    0.92  18.22  17.15    NaN    0.92  27.64
1  Ahmedabad  2015-01-02    NaN    NaN    0.97  15.69  16.46    NaN    0.97  24.55
2  Ahmedabad  2015-01-03    NaN    NaN   17.40  19.30  29.70    NaN   17.40  29.07
3  Ahmedabad  2015-01-04    NaN    NaN    1.70  18.48  17.97    NaN    1.70  18.59
4  Ahmedabad  2015-01-05    NaN    NaN   22.10  21.42  37.76    NaN   22.10  39.33

      O3  Benzene  Toluene  Xylene  AQI  AQI_Bucket
0  133.36    0.00    0.02    0.00    NaN      NaN
1   34.06    3.68    5.50    3.77    NaN      NaN
2   30.70    6.80   16.40    2.25    NaN      NaN
3   36.08    4.43   10.14    1.00    NaN      NaN
4   39.31    7.01   18.89    2.78    NaN      NaN

===== LAST FIVE ROWS =====
      City      Date  PM2.5  PM10    NO    NO2    NOx    NH3  \
29526  Visakhapatnam  2020-06-27   15.02  50.94    7.68  25.06  19.54  12.47
29527  Visakhapatnam  2020-06-28   24.38  74.09    3.42  26.06  16.53  11.99
29528  Visakhapatnam  2020-06-29   22.91  65.73    3.45  29.53  18.33  10.71
29529  Visakhapatnam  2020-06-30   16.64  49.97    4.05  29.26  18.80  10.03
29530  Visakhapatnam  2020-07-01   15.00  66.00    0.40  26.85  14.05   5.20
```

29529	0.52	9.84	28.30	0.00	0.00	0.00	54.0	Satisfactory
29530	0.59	2.10	17.05	NaN	NaN	NaN	50.0	Good

===== DATASET INFORMATION =====

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 29531 entries, 0 to 29530

Data columns (total 16 columns):

#	Column	Non-Null Count	Dtype
0	City	29531 non-null	object
1	Date	29531 non-null	object
2	PM2.5	24933 non-null	float64
3	PM10	18391 non-null	float64
4	NO	25949 non-null	float64
5	NO2	25946 non-null	float64
6	NOx	25346 non-null	float64
7	NH3	19203 non-null	float64
8	CO	27472 non-null	float64
9	SO2	25677 non-null	float64
10	O3	25509 non-null	float64
11	Benzene	23908 non-null	float64
12	Toluene	21490 non-null	float64
13	Xylene	11422 non-null	float64
14	AQI	24850 non-null	float64
15	AQI_Bucket	24850 non-null	object

dtypes: float64(13), object(3)

memory usage: 3.6+ MB

===== STATISTICAL SUMMARY =====

	City	Date	PM2.5	PM10	NO \
count	29531	29531	24933.000000	18391.000000	25949.000000
unique	26	2009	NaN	NaN	NaN
top	Ahmedabad	2020-06-26	NaN	NaN	NaN
freq	2009	26	NaN	NaN	NaN
mean	NaN	NaN	67.450578	118.127103	17.574730
std	NaN	NaN	64.661449	90.605110	22.785846
min	NaN	NaN	0.040000	0.010000	0.020000
25%	NaN	NaN	28.820000	56.255000	5.630000
50%	NaN	NaN	48.570000	95.680000	9.890000
75%	NaN	NaN	80.590000	149.745000	19.950000
max	NaN	NaN	949.990000	1000.000000	390.680000

	NO2	NOx	NH3	CO	SO2 \
count	25946.000000	25346.000000	19203.000000	27472.000000	25677.000000
unique	NaN	NaN	NaN	NaN	NaN
top	NaN	NaN	NaN	NaN	NaN
freq	NaN	NaN	NaN	NaN	NaN
mean	28.560659	32.309123	23.483476	2.248598	14.531977
std	24.474746	31.646011	25.684275	6.962884	18.133775
min	0.010000	0.000000	0.010000	0.000000	0.010000
25%	11.750000	12.820000	8.580000	0.510000	5.670000
50%	21.690000	23.520000	15.850000	0.890000	9.160000
75%	37.620000	40.127500	30.020000	1.450000	15.220000
max	362.210000	467.630000	352.890000	175.810000	193.860000

	O3	Benzene	Toluene	Xylene	AQI \
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	O3	Benzene	Toluene	Xylene	AQI \
count	25509.000000	23908.000000	21490.000000	11422.000000	24850.000000
unique	NaN	NaN	NaN	NaN	NaN
top	NaN	NaN	NaN	NaN	NaN
freq	NaN	NaN	NaN	NaN	NaN
mean	34.491430	3.280840	8.700972	3.070128	166.463581
std	21.694928	15.811136	19.969164	6.323247	140.696585
min	0.010000	0.000000	0.000000	0.000000	13.000000
25%	18.860000	0.120000	0.600000	0.140000	81.000000
50%	30.840000	1.070000	2.970000	0.980000	118.000000
75%	45.570000	3.080000	9.150000	3.350000	208.000000
max	257.730000	455.030000	454.850000	170.370000	2049.000000

	AQI_Bucket
count	24850
unique	6
top	Moderate
freq	8829
mean	NaN
std	NaN
min	NaN
25%	NaN
50%	NaN
75%	NaN
max	NaN

Dataset Shape: (29531, 16)

===== COLUMN NAMES =====

['City', 'Date', 'PM2.5', 'PM10', 'NO', 'NO2', 'NOx', 'NH3', 'CO', 'SO2', 'O3', 'Benzene', 'Toluene', 'Xylene', 'AQI', 'AQI_Bucket']

===== MISSING VALUES =====

City	0
Date	0
PM2.5	4598
PM10	11140
NO	3582
NO2	3585
NOx	4185
NH3	10328
CO	2059
SO2	3854
O3	4022
Benzene	5623
Toluene	8041
Xylene	18109
AQI	4681
AQI_Bucket	4681

dtype: int64

CSV file exported successfully!

Excel file exported successfully!

Result: